# Problem Set 3

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# Question 1: Spending and incumbent vote share

We want to see how the incumbent's vote share varies with the spending gap. We measure this gap as log incumbent spending minus log challenger spending (`difflog`).

(a) Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

**Answer:** First, we import the data and check the three variables.

```
# 1(a) Import the data and check the three variables.
incumbents <- read.csv("incumbents_subset.csv", row.names = 1)
incumbents <- incumbents[c("voteshare", "difflog", "presvote")]
print(dim(incumbents))
print(colSums(is.na(incumbents)))
summary(incumbents)
```

```
[1] 3193    3
voteshare   difflog  presvote
        0         0         0
   voteshare         difflog           presvote
 Min.   :0.3476   Min.   :-3.0601   Min.   :0.1931
 1st Qu.:0.5846   1st Qu.: 0.6734   1st Qu.:0.4695
 Median :0.6569   Median : 1.6106   Median :0.5478
 Mean   :0.6552   Mean   : 1.8278   Mean   :0.5512
 3rd Qu.:0.7233   3rd Qu.: 2.9857   3rd Qu.:0.6240
 Max.   :0.9930   Max.   : 5.8558   Max.   :0.9606
```

All 3,193 elections have complete data. The mean incumbent vote share is 0.6552, or 65.52%.

Then we use `lm()` to regress incumbent vote share on the spending gap:

```
# 1(a) Regress voteshare on difflog.
vote_spending <- lm(voteshare ~ difflog, data = incumbents)
summary(vote_spending)
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.579031   0.002251  257.19   <2e-16 ***
difflog     0.041666   0.000968   43.04   <2e-16 ***

Multiple R-squared:  0.3673,    Adjusted R-squared:  0.3671
```

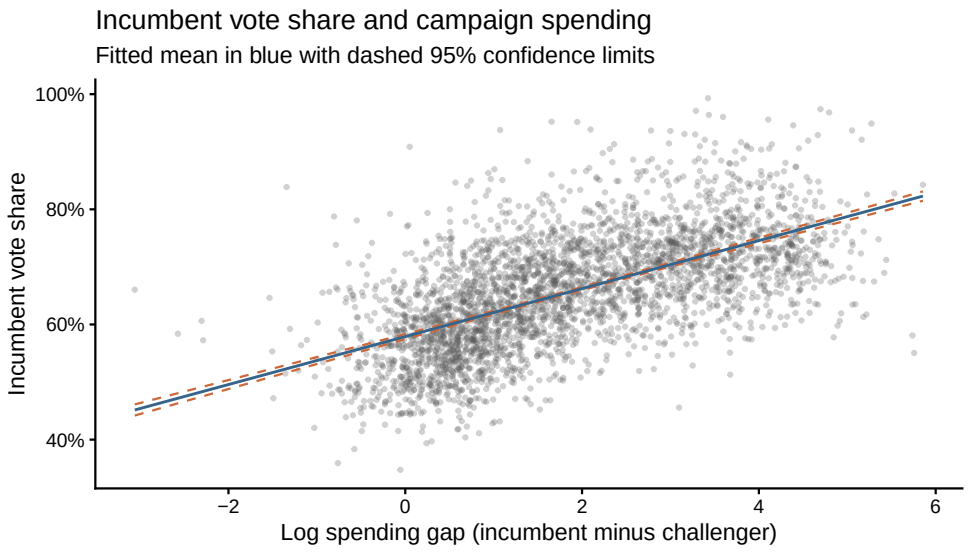
The spending gap explains 36.73% of the variation in incumbent vote share. The slope is positive, and we reject the null hypothesis of a zero slope at the 5% level (two-sided $p < 0.001$). Since spending was not randomly assigned, this is an association.

(b) Make a scatterplot of the two variables and add the regression line.

**Answer:** First, we calculate the fitted mean vote share and its 95% confidence limits over the observed range of spending gaps.

```
# 1(b) Calculate 95% confidence limits for the fitted mean.
library(ggplot2)
dir.create("Figures", showWarnings = FALSE)
spending_gap <- seq(min(incumbents$difflog), max(incumbents$difflog),
                    length.out = 200)
vote_grid <- data.frame(difflog = spending_gap)
vote_grid <- cbind(vote_grid, predict(vote_spending, vote_grid,
                              interval = "confidence", level = 0.95))
```

```
# 1(b) Plot the observations, fitted line and confidence limits.
vote_spending_plot <- ggplot(vote_grid, aes(x = difflog)) +
  geom_point(data = incumbents, aes(y = voteshare),
             size = 0.8, alpha = 0.28, colour = "grey35") +
  geom_line(aes(y = lwr), colour = "sienna3", linetype = "dashed") +
  geom_line(aes(y = upr), colour = "sienna3", linetype = "dashed") +
  geom_line(aes(y = fit), colour = "steelblue4", linewidth = 0.7) +
  scale_y_continuous(labels = scales::label_percent(accuracy = 1)) +
  labs(title = "Incumbent vote share and campaign spending",
       subtitle = "Fitted mean in blue with dashed 95% confidence limits",
       x = "Log spending gap (incumbent minus challenger)",
       y = "Incumbent vote share") +
  theme_classic(base_size = 12)
ggsave("Figures/Q1_vote_spending.pdf", vote_spending_plot,
       width = 7, height = 4)
```




Districts with the same spending gap can have quite different results. The dashed lines show uncertainty about the *mean* vote share, so individual elections can fall well outside them.

(c) Save the residuals of the model in a separate object.

**Answer:** We save the residuals and check the first election. A residual is the observed vote share minus the fitted vote share.

```
# 1(c) Save the residuals and check the first election.
vote_resid <- resid(vote_spending)
first_election <- c(observed = incumbents$voteshare[1],
                    fitted = unname(fitted(vote_spending)[1]),
                    residual = unname(vote_resid[1]))
print(round(first_election, 6))
```

```
observed    fitted  residual
0.602361  0.602784 -0.000423
```

The first incumbent received 60.24% of the vote, about 0.042 percentage points below the prediction. This explains the negative residual.

(d) Write the prediction equation.

**Answer:** We take the intercept and slope from the fitted model:

```
# 1(d) Get the coefficients for the prediction equation.
round(coef(vote_spending), 6)
```

```
(Intercept)     difflog
   0.579031    0.041666
```

Let $D_i$ be the log spending gap and $V_i$ the incumbent's vote share in election $i$. Using the rounded coefficients, the prediction equation is

$$\widehat{V}_i = 0.579031 + 0.041666\, D_i.$$

At equal spending ($D_i = 0$), the predicted incumbent vote share is 57.90%. A one-unit increase in the log spending gap is associated with 4.17 percentage points more incumbent vote share, on average.

# Question 2: Spending and presidential vote share

We now look at the same spending gap, but change the outcome to the district vote share of the presidential candidate from the incumbent's party (`presvote`).

(a) Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

**Answer:** We use `difflog` as in Question 1. This time we want to know whether the spending gap also lines up with support for the party's presidential candidate.

```
# 2(a) Regress presvote on difflog.
pres_spending <- lm(presvote ~ difflog, data = incumbents)
summary(pres_spending)
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.507583   0.003161  160.60   <2e-16 ***
difflog     0.023837   0.001359   17.54   <2e-16 ***

Multiple R-squared:  0.08795,   Adjusted R-squared:  0.08767
```

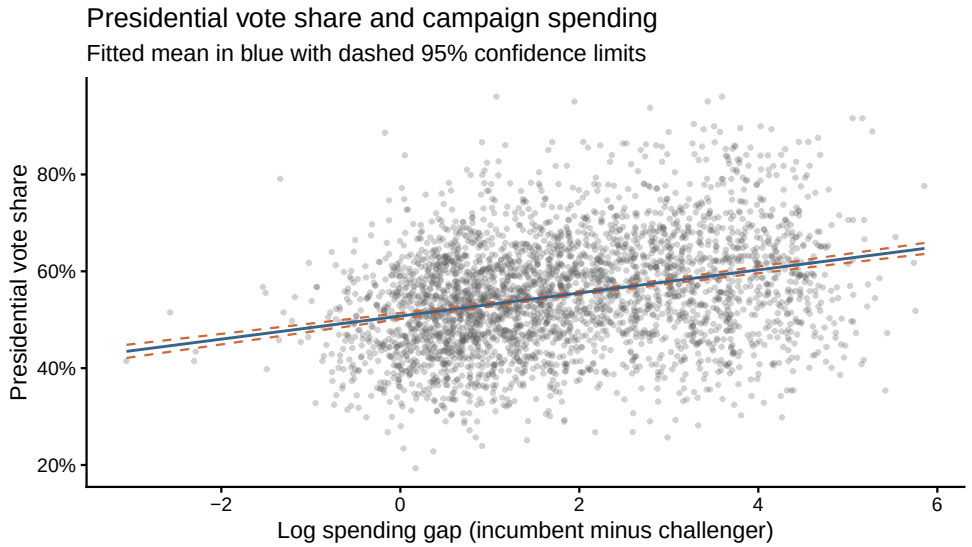
The estimated slope is positive and differs from zero at the 5% level ($p < 0.001$). The spending gap explains 8.80% of the variation in presidential vote share, compared with 36.73% in incumbent vote share in Question 1.

(b) Make a scatterplot of the two variables and add the regression line.

**Answer:** We put the spending gap on the horizontal axis and presidential vote share on the vertical axis. We calculate the 95% confidence limits from the model in part (a).

```
# 2(b) Calculate 95% confidence limits for the fitted mean.
pres_grid <- data.frame(difflog = spending_gap)
pres_grid <- cbind(pres_grid, predict(pres_spending, pres_grid,
                           interval = "confidence", level = 0.95))
```

```
# 2(b) Plot the observations, fitted line and confidence limits.
pres_spending_plot <- ggplot(pres_grid, aes(x = difflog)) +
  geom_point(data = incumbents, aes(y = presvote),
             size = 0.8, alpha = 0.28, colour = "grey35") +
  geom_line(aes(y = lwr), colour = "sienna3", linetype = "dashed") +
  geom_line(aes(y = upr), colour = "sienna3", linetype = "dashed") +
  geom_line(aes(y = fit), colour = "steelblue4", linewidth = 0.7) +
  scale_y_continuous(labels = scales::label_percent(accuracy = 1)) +
  labs(title = "Presidential vote share and campaign spending",
       subtitle = "Fitted mean in blue with dashed 95% confidence limits",
       x = "Log spending gap (incumbent minus challenger)",
       y = "Presidential vote share") +
  theme_classic(base_size = 12)
ggsave("Figures/Q2_pres_spending.pdf", pres_spending_plot,
       width = 7, height = 4)
```




The line slopes upward, but the points are more spread out than in Question 1. The spending gap tells us less about the presidential vote.

(c) Save the residuals of the model in a separate object.

**Answer:** We save the presidential residuals for Question 4 and check the first election:

```
# 2(c) Save the residuals and check the first election.
pres_resid <- resid(pres_spending)
first_election <- c(observed = incumbents$presvote[1],
                    fitted = unname(fitted(pres_spending)[1]),
                    residual = unname(pres_resid[1]))
print(round(first_election, 6))
```

```
observed   fitted residual
0.526778 0.521173 0.005606
```

The first residual is positive: the presidential candidate received 0.56 percentage points more than predicted.

(d) Write the prediction equation.

**Answer:** We can read the two coefficients from the model:

```
# 2(d) Get the coefficients for the prediction equation.
round(coef(pres_spending), 6)
```

```
(Intercept)     difflog
   0.507583    0.023837
```

Let $P_i$ be the presidential vote share. Then

$$\widehat{P}_i = 0.507583 + 0.023837\, D_i.$$

At equal spending, the predicted share is 50.76%. Each additional log unit of the gap corresponds to 2.38 percentage points more presidential vote share.

# Question 3: The two vote shares

How is support for the presidential candidate of the incumbent's party related to the incumbent's own vote share?

(a) Run a regression where the outcome variable is `voteshare` and the explanatory variable is `presvote`.

**Answer:** We use the incumbent's vote share as the outcome and the presidential vote share as the explanatory variable. For now, we do not control for spending.

```
# 3(a) Regress voteshare on presvote.
vote_pres <- lm(voteshare ~ presvote, data = incumbents)
summary(vote_pres)
```

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.441330   0.007599   58.08   <2e-16 ***
presvote    0.388018   0.013493   28.76   <2e-16 ***

Multiple R-squared:  0.2058,    Adjusted R-squared:  0.2056
```

Presidential vote share explains 20.58% of the variation in incumbent vote share. The positive slope differs from zero at the 5% level ($p < 0.001$): districts that give the party more support in the presidential race also tend to give its incumbent a larger share of the vote.

(b) Make a scatterplot of the two variables and add the regression line.

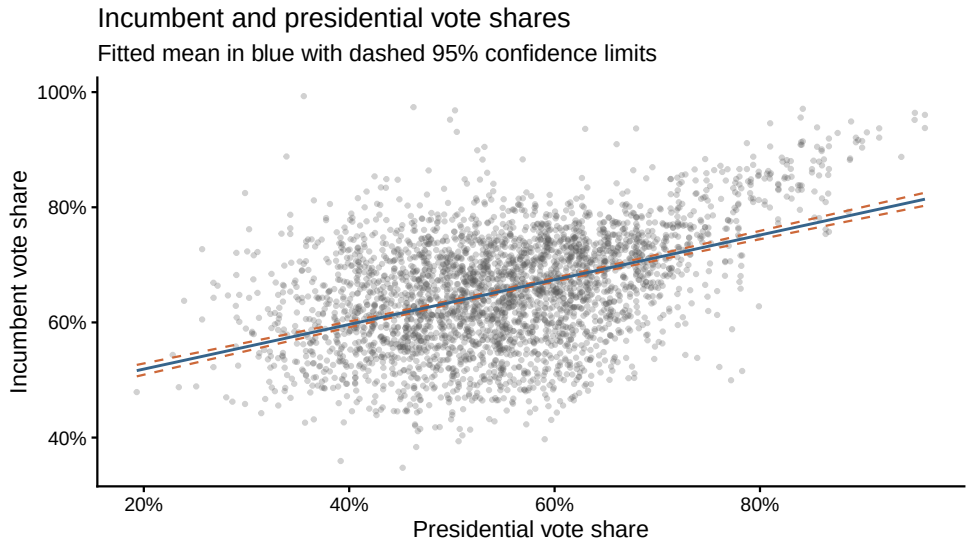
**Answer:** We calculate the fitted incumbent vote share and its 95% confidence limits at different values of `presvote`.

```
# 3(b) Calculate 95% confidence limits for the fitted mean.
pres_share <- seq(min(incumbents$presvote), max(incumbents$presvote),
                  length.out = 200)
support_grid <- data.frame(presvote = pres_share)
support_grid <- cbind(support_grid, predict(vote_pres, support_grid,
                                  interval = "confidence", level = 0.95))
```

```
# 3(b) Plot the observations, fitted line and confidence limits.
vote_pres_plot <- ggplot(support_grid, aes(x = presvote)) +
  geom_point(data = incumbents, aes(y = voteshare),
             size = 0.8, alpha = 0.28, colour = "grey35") +
  geom_line(aes(y = lwr), colour = "sienna3", linetype = "dashed") +
  geom_line(aes(y = upr), colour = "sienna3", linetype = "dashed") +
  geom_line(aes(y = fit), colour = "steelblue4", linewidth = 0.7)
```

Finally, we label both axes as percentages and save the plot.

```
# Label the axes and save the plot.
vote_pres_plot <- vote_pres_plot +
  scale_x_continuous(labels = scales::label_percent(accuracy = 1)) +
  scale_y_continuous(labels = scales::label_percent(accuracy = 1)) +
  labs(title = "Incumbent and presidential vote shares",
       subtitle = "Fitted mean in blue with dashed 95% confidence limits",
       x = "Presidential vote share",
       y = "Incumbent vote share") +
  theme_classic(base_size = 12)
ggsave("Figures/Q3_vote_pres.pdf", vote_pres_plot,
       width = 7, height = 4)
```




The fitted line rises from left to right. Even where the presidential candidate received about half the vote, some incumbents received less than half and others considerably more.

(c) Write the prediction equation.

**Answer:** We use the estimated intercept and the coefficient on `presvote`:

```
# 3(c) Get the coefficients for the prediction equation.
round(coef(vote_pres), 6)
```

```
(Intercept)    presvote
   0.441330    0.388018
```

The prediction equation is

$$\widehat{V}_i = 0.441330 + 0.388018\, P_i.$$

For example, a change in presidential vote share from 50% to 60% gives $0.388018(0.60 - 0.50) = 0.03880$, or 3.88 percentage points more incumbent vote share.

# Question 4: The added-variable plot

In Questions 1(c) and 2(c), we saved the part of each vote share left unexplained by the spending gap. Now we compare those two residuals.

(a) Run a regression where the outcome variable is the residual from Question 1 and the explanatory variable is the residual from Question 2.

**Answer:** For each election, we use the incumbent's residual as the outcome and the presidential candidate's residual as the explanatory variable.

```
# 4(a) Regress vote_resid on pres_resid.
residual_data <- data.frame(vote_resid, pres_resid)
resid_model <- lm(vote_resid ~ pres_resid, data = residual_data)
summary(resid_model)
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.942e-18  1.299e-03    0.00        1
pres_resid   2.569e-01  1.176e-02   21.84   <2e-16 ***

Multiple R-squared:   0.13, Adjusted R-squared:  0.1298
```

The intercept is effectively zero because both sets of residuals have mean zero. The slope is positive and differs from zero at the 5% level ($p < 0.001$). The presidential residual explains 13.00% of the variation *left over* in incumbent vote share after accounting for spending.

(b) Make a scatterplot of the two residuals and add the regression line.

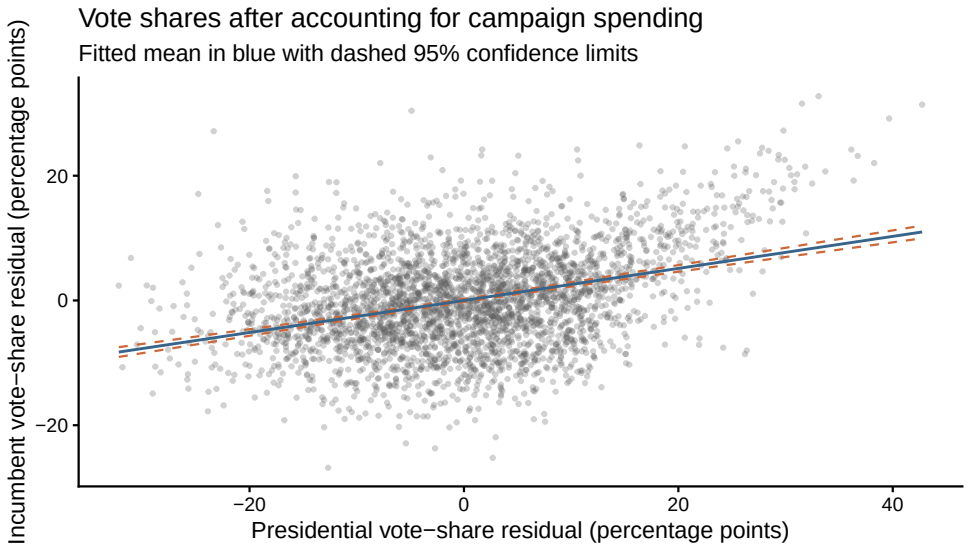
**Answer:** We use the model from part (a) to calculate the fitted incumbent residual and its 95% confidence limits across the observed presidential residuals.

```
# 4(b) Calculate 95% confidence limits for the fitted mean.
pres_gap <- seq(min(residual_data$pres_resid), max(residual_data$pres_resid),
                length.out = 200)
resid_grid <- data.frame(pres_resid = pres_gap)
resid_grid <- cbind(resid_grid, predict(resid_model, resid_grid,
                                interval = "confidence", level = 0.95))
```

```
# 4(b) Plot the observations, fitted line and confidence limits.
resid_plot <- ggplot(resid_grid, aes(x = pres_resid)) +
  geom_point(data = residual_data, aes(y = vote_resid),
             size = 0.8, alpha = 0.28, colour = "grey35") +
  geom_line(aes(y = lwr), colour = "sienna3", linetype = "dashed") +
  geom_line(aes(y = upr), colour = "sienna3", linetype = "dashed") +
  geom_line(aes(y = fit), colour = "steelblue4", linewidth = 0.7)
```

To read the residuals in percentage points, we multiply the axis labels by 100.

```
# Label the axes and save the plot.
resid_plot <- resid_plot +
  scale_x_continuous(labels = scales::label_number(scale = 100)) +
  scale_y_continuous(labels = scales::label_number(scale = 100)) +
  labs(title = "Vote shares after accounting for campaign spending",
       subtitle = "Fitted mean in blue with dashed 95% confidence limits",
       x = "Presidential vote−share residual (percentage points)",
       y = "Incumbent vote−share residual (percentage points)") +
  theme_classic(base_size = 12)
ggsave("Figures/Q4_added_variable.pdf", resid_plot,
       width = 7, height = 4)
```




The line slopes upward. In districts in the upper right, both candidates did better than the spending gap predicted.

(c) Write the prediction equation.

**Answer:** We take the coefficients from the residual regression:

```
# 4(c) Get the coefficients for the prediction equation.
round(coef(resid_model), 6)
```

```
(Intercept)  pres_resid
   0.000000    0.256877
```

Let $r_{Vi}$ and $r_{Pi}$ be the incumbent and presidential residuals. The intercept rounds to zero, so the prediction equation is

$$\widehat{r}_{Vi} = 0.256877\, r_{Pi}.$$

When the presidential candidate does one percentage point better than predicted from spending, the incumbent is predicted to do about 0.26 points better as well.

# Question 5: The multiple regression

Finally, we include presidential vote share and the spending gap in the same model for incumbent vote share.

(a) Run a regression where the outcome variable is `voteshare` and the explanatory variables are `difflog` and `presvote`.

**Answer:** We include both variables so that we can interpret each coefficient while holding the other variable fixed.

```
# 5(a) Regress incumbent vote share on both predictors.
full_model <- lm(voteshare ~ difflog + presvote, data = incumbents)
summary(full_model)
```

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.4486442  0.0063297   70.88   <2e-16 ***
difflog     0.0355431  0.0009455   37.59   <2e-16 ***
presvote    0.2568770  0.0117637   21.84   <2e-16 ***

Multiple R-squared:  0.4496,    Adjusted R-squared:  0.4493
```

Both slopes are positive and differ from zero at the 5% level ($p < 0.001$ for each). Together, the two variables explain 44.96% of the variation in incumbent vote share.

(b) Write the prediction equation.

**Answer:** We use the intercept and both slopes:

```
# 5(b) Get the coefficients for the prediction equation.
round(coef(full_model), 6)
```

```
(Intercept)     difflog    presvote
   0.448644    0.035543    0.256877
```

Therefore, the prediction equation is

$$\widehat{V}_i = 0.448644 + 0.035543\, D_i + 0.256877\, P_i.$$

At the same presidential vote share, a one-unit increase in the log spending gap is associated with 3.55 percentage points more incumbent vote share. At the same spending gap, a 10-percentage-point increase in presidential vote share is associated with 2.57 percentage points more incumbent vote share, compared with 3.88 before adjusting for spending in Question 3.

(c) What is identical to the output in Question 4? Why?

**Answer:** First, we compare the two slopes directly in R. Then we check whether the two models leave the same residuals for each election:

```r
# 5(c) Compare the slopes and residuals from Questions 4 and 5.
slopes <- c(Q4 = unname(coef(resid_model)["pres_resid"]),
            Q5 = unname(coef(full_model)["presvote"]))
print(slopes, digits = 10)
print(all.equal(unname(resid(resid_model)), unname(resid(full_model))))
```

```
          Q4           Q5
0.2568770127 0.2568770127
[1] TRUE
```

The slope on presidential vote share in Question 5 and the slope on the presidential residual in Question 4 are both 0.2568770127. The residuals also match, up to numerical precision.

In Question 4, we first removed from each vote share the part predicted by the spending gap and then compared what was left. In Question 5, holding the spending gap fixed makes the same comparison. That is why we get the same slope (the Frisch–Waugh–Lovell result).

The two models do not explain the same fraction of their outcomes: Question 4 explains 13.00% of the variation *remaining after spending*, while Question 5 explains 44.96% of the variation in incumbent vote share itself. Question 5 also has 3,190 residual degrees of freedom, one fewer than Question 4.